RUNGTAINTERNATIONALSKILLSUNIVERSITY
SCHOOLOFCOMPUTERSCIENCE AND ENGINEERING

Bachelor of Technology Computer Science and Engineering

PSNo.	34
ojectTitle	**Tic-Tac-Toe**
roblem tement	Developatwo-playerconsoleTic-Tac-Toegameona3×3boardwhereplayerstaketurns,theboardisdisplayed after every move, and the winner or a draw is announced. **Language:**Python3 **Concepts:**Lists,Functions, Conditions,GameLogic **Note:**Positionsarenumbered1to9(1-2-3toprow,4-5-6middlerow,7-8-9bottomrow).PlayerXalwaysplays first.

INPUT	EXPECTEDOUTPUT
TestCase1:NormalInput	
Moves:X:1,O:4,X:2,O:5,X:3	Player X wins! Winningline:1-2-3
TestCase2:Boundary/MinimumInput	
Moves:X:1,O:2,X:3,O:5,X:4,O:6,X:8,O:7,X:9	Boardisfullafter9moves. It's a draw! No moves left.
TestCase3:InvalidInput	
(a) Position:10 (b) Position:abc (c) Xplays1,thenOplays1	(a) Invalidposition.Enteranumberfrom1to9. (b) Invalidposition.Enteranumberfrom1to9. (c) Position1isalreadytaken.Chooseanotherposition. (turn does not change)
TestCase4:Different ValidScenario	
Moves:X:1,O:4,X:2,O:5,X:9,O:6 Playagain? n	Player O wins! Winningline:4-5-6 Thanksforplaying!

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1. WritethesolutioninPython3(one.pyfile) and run itfromtheconsole.
2. Storetheboard in alistof9 cellsanddisplay itaftereverymove.
3. Writefunctionstodisplaytheboard,takeamove,checkforawin andcheckfora draw.
4. Checkall8winninglines(3rows,3 columns,2 diagonals)using conditions.
5. Validateeverymoveand askPlayagain?(y/n)'attheend.

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1. Understandtheproblemstatement anddefinetheproblemclearly.
2. Developthesolutionusingproperlogicandprogramming concepts.
3. Implementallrequiredfeaturesandpassthegiventestcases.
4. Testanddebugtheprojecttoensurecorrect output.
5. Preparethedocumentation/reportanduploadthecodeandreportto GitHub.
6. Completeandsubmittheprojectwithin thegiven deadline.